Contents

1 Results — System Evaluation **2**
1.1 Methodology . . . 2
1.2 Instrument selection rationale . . . 2
1.3 ML Test Score . . . 2
1.4 Reproducibility . . . 3
1.5 Maturity-model placement . . . 4
1.6 Threats to validity . . . 5
1.7 Known gaps and future work (system side) . . . 5
1.8 Summary . . . 5

Chapter 1

Results — System Evaluation

1.1 Methodology

- This chapter evaluates the *system*; forecast accuracy lives in Chapter **??**.
- Frame as a single-case study [4]: source code, git/CI history, ClickHouse experiment tracker, short targeted experiments.
- Honesty boundary: figures are indicative over the study window, not long-run production claims.
- Domain precedent: matches how MLOps platforms for electricity-price forecasting are evaluated [5].
- Explicitly not claimed: controlled comparison against another platform.

1.2 Instrument selection rationale

- Open: Zarour's SLR [6] catalogues nine practices but no single quantitative score; we therefore compose two instruments, each addressing a measurable face of the system.
- Map the two instruments to the practices they evidence:
  - ML Test Score (§1.3) → aggregate evidence for P1, P2, P3, P4, P6, P7, P8 (rubric items cover them).
  - Reproducibility (§1.4) → direct evidence for P3.
- One sentence on what is *not* instrumented quantitatively and why: P5 (Collaboration) — qualitative by nature, handled in System Implementation Ch. **??**.

1.3 ML Test Score

- Rubric origin [1]: 28 tests across four sections, each scored $\{0, 0.5, 1\}$; final = *min* of the four section sums.
- Min-of-sections rationale: the weakest section bounds the result and surfaces where investment is needed; rewards balance, not breadth.

- Binding section: Data tests (2.5 / 7).

Table 1.1: ML Test Score self-assessment. Per-item evidence in `Report/metrics/ml_test_score.md`.

Section (max 7)	**Sum**
Data tests	2.5
Model development tests	3.0
ML infrastructure tests	4.0
Monitoring tests	4.0
ML Test Score (min)	**2.5**

- What is scored: input schema enforcement (`enforce_schema`); a tested feature-transformer pipeline.
- What is missing: formal feature-benefit, feature-cost, privacy-controls analyses; no data-quality CI gate.
- Future work pointer: §1.7.
- *Optional rescore decision (your call):* S1-2 (Lasso is feature selection) and S1-4 (Lago-grounded feature set) are arguably 0.5 each — would lift Data to 3.0 or 3.5 and the headline to 3.0. State and defend, or leave conservative.

1.4 Reproducibility

- Two senses, measured separately: *provenance coverage* (a percentage) and *demonstration* (a worked re-run).
- Score framing follows [3]; ACM Artifact Review and Badging gives the badge mapping.

Provenance coverage

Table 1.2: Provenance triple coverage across 15 live runs in the window.

Provenance component	**Coverage**
`git_sha`	66.7 % (10 / 15)
`data_start_utc`	100.0 % (15 / 15)
`data_end_utc`	100.0 % (15 / 15)
`model_artifact_id`	100.0 % (15 / 15)
Full triple (all four)	**66.7 % (10 / 15)**

- Shortfall is entirely on `git_sha`; data window and artefact are 100 % from day one.
- Cause: 5 of 15 runs pre-date the CD pipeline baking `GIT_SHA` into the production image.
- From the first cd-deployed run onward, all four components are populated.

Re-run demonstration

- Only inputs supplied: the run's committed config + target date; stored prediction loaded from `algo.experiment_predictions`.

Table 1.3: Reproducibility re-run: regenerated from logged provenance, compared against the stored prediction.

Field	Value
Config	`dk1_lear`
Target date	2026-03-22
Delivery periods	24
Max absolute difference	7×10^{-6} EUR/MWh
Verdict	numerically identical
Command	`uv run python -m scripts.report.reproduce_run -config dk1_lear -date 2026-03-22`

- 7×10^{-6} EUR/MWh is five orders of magnitude below any economically meaningful price unit — solver-level floating-point tolerance.
- Deterministic LEAR used for the headline demo; DNN / Transformer carry seed and ensemble stochasticity (note as caveat).
- ACM Artifact Badging [3]: *Available, Functional, Reusable* by self-review; *Reproduced* not claimed (requires external reviewer).

1.5 Maturity-model placement

- Cross-reference the qualitative practice-by-practice table in System Implementation (Ch. **??**, Table **??**) — do not duplicate it here.
- Open the section with the methodological question: *which* maturity model should the per-practice levels be assigned against, and why?

Table 1.4: The five MLOps maturity models reviewed by Zarour et al. [6], Table 7. All trace back to CMM/CMMI lineage. The system is positioned against the **Google 3-level** model (chosen below).

Model	Levels	Top level description
Google	3 (0–2)	CI/CD pipeline automation
Microsoft	5 (0–4)	Full MLOps automated operations
GigaOm	5 (0–4)	Comprehensive MLOps pipeline supporting frequent model updates
John, Olsson & Bosch (2021)	4 (0–3)	Fully-automated model monitoring
Mateo-Casalí et al. (2022)	5 (0–4)	Optimization (CMMI mapping)

- Justify the choice of Google in three bullets:
  - **Granularity matches scope**: the system is a single-host MVP; a 3-level scheme avoids over-claiming distinctions between Microsoft's Levels 2–4 (Automated Training / Auto Deployment / Full Automation) that a single deployment target cannot meaningfully differentiate.
  - **Top level is plausibly attainable**: Google Level 2 ("CI/CD pipeline automation") maps cleanly to the realised platform; Microsoft Level 4 ("Full MLOps automated operations") would require multi-environment claims the platform does not make.

  - **Open and widely cited**: Google's whitepaper is publicly accessible [2] and is the most cited industry maturity reference in the Zarour SLR, giving the assignment an external touchpoint reviewers can check.

- Disclose alternatives honestly: a Microsoft, GigaOm, John et al., or Mateo-Casalí scoring is possible and would yield different per-practice numbers; we report Google levels in Table **??** for these reasons and note that none of the five models has empirical validation comparable to a peer-reviewed instrument.
- Limitation: all five models are descriptive, vendor- or academia-defined ordinals — they do not produce a quantitative aggregate; they support *positioning*, not *measurement*. The instruments in §1.3–1.4 carry the measurement load.

1.6 Threats to validity

- Frame per [4]; one bullet per threat type.
- **Construct**: do the chosen scores measure what we claim? ML Test Score is a self-administered rubric; per-item evidence in `ml_test_score.md` makes claims checkable.
- **Internal**: are observed properties causally attributable to the platform? Reproducibility speaks directly from artefacts.
- **External**: single-case design, analytic generalisation only — findings are not statistical claims about MLOps platforms in general.
- **Reliability**: would another evaluator score the same? Scoring rules in §1.3 are explicit; evidence is recorded per item.
- Alignment with Zarour's own SLR validity discussion (its §4.5): the threats we name parallel internal/external/construct in their paper [6].

1.7 Known gaps and future work (system side)

- **Data tests** — formalise feature-benefit / feature-cost / privacy analyses; add a data-quality CI gate. Drives the binding ML Test Score section.
- **Reproducibility extension** — DNN / Transformer re-run with controlled seeds; currently only deterministic LEAR is demonstrated.
- **Provenance backfill** — 5 pre-CD runs lack `git_sha`; either backfill from log timestamps or document the gap explicitly.
- Cross-reference modelling-side future work in Chapter **??**.

1.8 Summary

- One-paragraph summary: the chapter's qualitative "Realised" labels are backed by an ML Test Score of 2.5 and a provenance-triple coverage of 66.7 % supported by a 7×10^{-6} EUR/MWh re-run.
- Maturity position: Google Level 2 across most practices, Level 1 for P5–P7 and P9 (see Ch. **??**, Table **??**).
- Cross-reference Chapter **??** (model accuracy) and Chapter **??** (qualitative practice-by-practice).

Bibliography

[1] Eric Breck, Shanqing Cai, Eric Nielsen, Michael Salib, and D. Sculley. The ML test score: A rubric for ML production readiness and technical debt reduction. In *2017 IEEE International Conference on Big Data (Big Data)*, pages 1123–1132, 2017.

[2] Google Cloud. MLOps: Continuous delivery and automation pipelines in machine learning, 2024. Google Cloud Architecture Center whitepaper. Defines a three-level MLOps maturity model (Level 0 manual, Level 1 ML pipeline automation, Level 2 CI/CD pipeline automation).

[3] Odd Erik Gundersen and Sigbjørn Kjensmo. State of the art: Reproducibility in artificial intelligence. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.

[4] Per Runeson and Martin Höst. Guidelines for conducting and reporting case study research in software engineering. *Empirical Software Engineering*, 14(2):131–164, 2009.

[5] Rakshith Subramanya, Seppo Sierla, and Valeriy Vyatkin. From DevOps to MLOps: Overview and application to electricity market forecasting. *Applied Sciences*, 12(19):9851, 2022.

[6] Mohammad Zarour, Hussein Alzabut, and Khalid T. Al-Sarayreh. MLOps best practices, challenges and maturity models: A systematic literature review. *Information and Software Technology*, 183:107733, 2025.